

Playing the Monograph: Retrieval-Augmented Generation for Interactive Ethnography

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Abstract

Games are rarely used to engage with complex ethnographic texts, limiting how broader audiences encounter foundational works in anthropology. This work introduces our game Malinowski's Lens, an AI-native educational game that reimagines Bronisław Malinowski's *Argonauts of the Western Pacific* (1922) as an interactive fieldwork experience. Indigenous characters are rendered as silhouettes while Malinowski appears in detail, prompting reflection on longstanding issues of authorship and cultural representation in ethnographic practice. By tightly coupling generative AI and curated retrieval, our work investigates how AI-native game design can transform dense academic texts into accessible interactive experiences, encouraging players to develop curiosity about the specific monograph, its local histories, and the discipline of anthropology more broadly. We contribute design insights for using retrieval-augmented generative AI to transform dense anthropological texts into exploratory, ethically reflective interactive systems, highlighting how AI-native game mechanics can support interpretive learning without replacing scholarly reading.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Human computer interaction (HCI)**.

Keywords

Interactive Storytelling, Generative AI, Human-Computer Interaction, LLM, Educational Gaming

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1 Introduction

Recent advances in generative AI have drawn attention to the expanding role of large language models (LLMs) in creative production, including creative writing, storytelling, and theatrical generation, as well as emerging applications in game design [1, 6, 15, 16, 20]. Within games research, prior work has articulated multiple roles for LLMs, such as Player, Non-Player Character, Player Assistant, Commentator, Analyst, and Game Master, highlighting their potential to structure interactive, narrative-driven experiences [4]. However, less attention has been paid to how generative AI might support games grounded in academic texts, particularly in the humanities and social sciences.

In this work, we explore how generative AI can support the design of an anthropological game: an interactive system that invites players to engage with cultural material through the interpretive stance of a fieldworker. Rather than positioning such a system as a replacement for ethnographic reading, our aim is exploratory. We are interested in how an AI-driven game might function as a point of entry into anthropological thinking by sparking curiosity, supporting initial comprehension of key concepts, and motivating further engagement with primary texts. We suggest that, when framed in this way, generative AI can support learning experiences that complement rather than supplant traditional scholarly practices.

We present Malinowski's Lens, an AI-native educational game that reimagines Bronisław Malinowski's *Argonauts of the Western Pacific* [12] as an interactive learning environment. The system integrates two generative AI components: a Retrieval-Augmented Generation (RAG) pipeline that produces narrative sequences grounded in the source text, and a visual generation pipeline based on DALL-E 3 that responds to player actions while maintaining a consistent, VGA-inspired aesthetic. Together, these systems enable real-time



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narrative and visual feedback while anchoring gameplay in ethnographic material.

Players inhabit the role of Malinowski during his 1915–1918 fieldwork in the Trobriand Islands. Each player action prompts narrative continuations and visual updates, with the RAG system ensuring coherence with the ethnographic record. Rather than offering a comprehensive retelling of the book, the game selectively surfaces concepts, scenes, and tensions from the text, encouraging players to seek additional context and clarification through further reading.

Gameplay unfolds across two phases. In the fieldwork phase, players explore AI-generated cultural scenarios, interact with characters, and collect ethnographic observations. In the academic defense phase, players articulate and justify their interpretations to a virtual colleague, encouraging reflection and synthesis. To address concerns around cultural representation, the visual design abstracts depictions of Indigenous Trobriand Islanders as silhouettes while rendering Malinowski in greater detail. This design choice responds to ongoing discussions around representation and bias in AI-generated imagery [5, 7] and foregrounds the interpretive and political dimensions of ethnographic depiction.

Our work contributes (1) the design of an AI-native educational game that operationalizes ethnographic interpretation through play, (2) a RAG-based pipeline for grounding generative narrative interaction in canonical anthropological texts, and (3) design reflections on representational ethics when using generative imagery in cross-cultural contexts.

2 Related Work

Research in human–computer interaction and game studies has explored computational approaches to interactive storytelling that balance player agency with authored structure. Early systems such as *Faade* [13] and *Prom Week* [14] used rule-based social simulation and narrative planning to support responsive dramatic experiences. More recently, large language models have enabled open-ended improvisation and emergent dialogue in systems such as *AI Dungeon* [8], though prior work has identified challenges related to narrative drift, thematic incoherence, and weakened authorial intent under unconstrained generation.

In response, researchers have explored hybrid and constrained generative systems that embed language models within structured mechanics or co-creative frameworks. Co-creative storytelling systems such as *1001 Nights* [3, 18] guide generative output toward specific narrative forms, while reflective serious games such as *Cracking Aegis* [2] align generative AI with educational and ethical goals. Parallel work in educational HCI has adopted Retrieval-Augmented Generation (RAG) to ground LLM outputs in authoritative sources, improving transparency and trust in tutoring and feedback systems [11, 17, 19].

Our work addresses this gap by presenting an AI-native educational game that integrates RAG-based narrative generation with a fixed ethnographic corpus as its authoritative substrate. Rather than inventing a fictional world, the system mediates player access to an existing one as articulated by the anthropologist, positioning generative AI as an epistemic mediator that supports interpretive learning, disciplinary enculturation, and ethical reflection.

3 System Design and Implementation

We describe ‘Malinowski’s Lens’, including its game design, user interface and technical implementation.

3.1 Game Design

3.1.1 The Player’s Journey. The player begins by establishing a field camp on the Trobriand Islands, where the LLM generates scenarios inspired by Malinowski’s *Argonauts of the Western Pacific* [12]. Each turn presents a scene description, an accompanying image, three AI-generated response options, and a custom input field for open-ended exploration. After selecting or entering a response, a loading screen appears featuring a quotation from Malinowski, during which players can collect randomly appearing local expressions that reveal their meaning when clicked. Following a short loading period (40–50 seconds), the next scene is generated. During this interactive fieldwork, players investigate political, economic, and religious aspects of Trobriand society while collecting cultural items (artifacts, expressions and practices) which populate a dedicated Cultural Inventory. Once four cultural artefacts are collected, the fieldwork phase ends and the player returns to their university base for an academic challenge. Before entering this Academic Defense, players may review all prior scenes through the day counter. The Defense consists of a ten-question multiple-choice quiz assessing retention of key concepts encountered during play, after which a scoreboard provides immediate performance feedback¹.

3.1.2 Ethnographic Collection: Discovering Cultural Treasures. While exploring the Trobriand Islands, players engage with two primary activities. First, they navigate the political, economic, and religious dimensions of island life through scenario-based narratives derived from Malinowski’s “Argonauts of the Western Pacific” via the RAG-system. Second, they must identify and collect significant cultural elements such as tangible objects, conceptual insights and local expressions. These ethnographic treasures are automatically identified by a parser that analyzes the AI-generated content. When encountered, these cultural elements are added to the player’s inventory. This collection mechanic structures ethnographic learning through goal-oriented discovery, reinforcing key concepts via progression rather than exposition.

3.1.3 Dynamic Environment Design: Responsive Visuals with Ethical Considerations. To deepen immersion and engagement, we designed a dynamic environment system that generates responsive visuals in real time. Narrative scenarios, produced via the RAG system, are distilled into concise prompts for DALL-E 3, which in turn generates VGA-style background images. This design operationalizes principles of emotional game design [9] by aligning visual feedback directly with narrative progression, thereby reinforcing player–story connection.

Our choice of a consistent early-1990s VGA adventure game aesthetic was deliberate: it provides a cohesive visual identity, evokes nostalgia associated with exploratory gameplay, and differentiates the educational experience from contemporary photorealistic genres. Importantly, our visual representation strategy addressed the

¹A video demonstration of 4 minutes of gameplay as well as the quiz is available at [dropbox](#), anonymized to protect participant privacy.

ethical complexities of depicting indigenous communities. To mitigate risks of misrepresentation by Western-trained AI models, we rendered Trobriand Islanders as silhouettes while presenting Malinowski himself in detail. This technique echoes initiatives in art such as Silvia Kolbowski’s explorations of postcolonial shadows² but also draws from LaPensée [10] research on indigenous representation in digital media, which emphasizes respectful portrayal while acknowledging the inherent complexities of cross-cultural depiction. Preliminary playtests (N=14) indicated that players perceived this visual strategy as appropriate and fitting for the game’s specific anthropological context, and that it positively shaped their sense of immersion and authenticity. Importantly, we frame this silhouette-based representation not as a solution to representational ethics, but as a deliberate provocation that makes visible the asymmetries of ethnographic authorship. This approach suggests a broader design strategy for AI-generated cultural content: foregrounding absence and abstraction as a means of encouraging critical reflection rather than simulated authenticity.

3.1.4 Academic Defense: From Fieldwork to Scholarly Scrutiny. The second phase of the game, the Academic Defense, models the process of defending fieldwork in scholarly contexts. Once players have gathered four cultural artefacts, they enter a quiz-based dialogue where an AI agent plays the role of a critical academic peer. The system generates ten multiple-choice questions: one quotes Malinowski’s *Argonauts*, one addresses the theoretical basis of the game, three test vocabulary encountered during pauses, two focus on collected artifacts, and the remaining three probe narrative comprehension.

This mechanic creates a payoff structure for the fieldwork phase by requiring players to actively demonstrate retention and interpretation. It echoes existing platforms such as Kahoot! [21], but differs in its dynamic tailoring of questions to player-generated gameplay histories. The approach situates assessment not as an external add-on but as an embedded continuation of the anthropological journey.

3.2 User Interface



Figure 1: UI Interface

²Silvia Kolbowski is noted for her use of shadows and the deliberate absence of human figures. In works such as *Monumental Prop/Portions*, she removes the body from the image, leaving only cropped, expressive shadows (see <https://www.printedmatter.org/catalog/8949/>)

The user interface (UI) of Malinowski’s *Lens* is designed to balance narrative immersion with interactive learning, drawing inspiration from early text-adventure games while integrating affordances of modern educational technology. The screen is divided into two primary zones (Fig. 1): a left-side narrative space and a right-side interaction panel. The left panel features a pixel-based scene visualization, which dynamically changes based on the player’s choices. Below this image, a contextual text box presents a brief scene description grounded in retrieved ethnographic content. This visual-verbal pairing aims to support multimodal learning and narrative immersion, reinforcing the connection between player decisions and cultural context.

The right panel serves as the primary interaction zone. It presents four response options: three AI-generated choices and one open-ended input field that allows players to type their own response. This hybrid model supports both structured learning and free-form exploration, inviting players to co-construct meaning through natural language interaction. Beneath the interaction panel, additional UI elements track player progress. These include a cultural inventory (listing discovered insights and artifacts), a day counter, and real-time metrics indicating the number of collected expressions and cultural items. These components function both as feedback mechanisms and as learning reinforcements, visually marking progress through the gameplay.

3.3 Technical Architecture of the Interactive Storytelling System

The technical foundation of Malinowski’s *Lens* was developed to meet three core requirements: (1) ensuring fidelity to ethnographic source material, (2) enabling dynamic and adaptive gameplay, and (3) maintaining extensibility for future learning contexts. To address these goals, we implemented a modular Retrieval-Augmented Generation pipeline integrated with a layered application architecture (Fig. 2). At the knowledge retrieval layer, the system ingests Malinowski’s *Argonauts of the Western Pacific* (1922) [12]. The text is pre-processed into high-dimensional embeddings using the OpenAI text-embedding-3-small model³, and stored in a ChromaDB vector database for efficient retrieval. When players interact with the game, the retrieval engine queries ChromaDB⁴ to surface semantically relevant excerpts. This architecture constrains narrative generation to authentic ethnographic passages, reducing hallucinations and ensuring cultural and textual fidelity.

The narrative generation layer integrates retrieval with generative modeling. Retrieved passages are dynamically composed with structured prompts orchestrated through LangChain⁵, and then passed to an LLM (GPT-4o). This setup ensures that generated content remains contextually grounded while preserving the flexibility required for interactive storytelling. Prompt templates explicitly encode pedagogical goals, guiding the model to produce both culturally accurate and educationally coherent scenarios.

³<https://platform.openai.com/docs/models/text-embedding-3-small>

⁴<https://www.trychroma.com>

⁵<https://www.langchain.com>

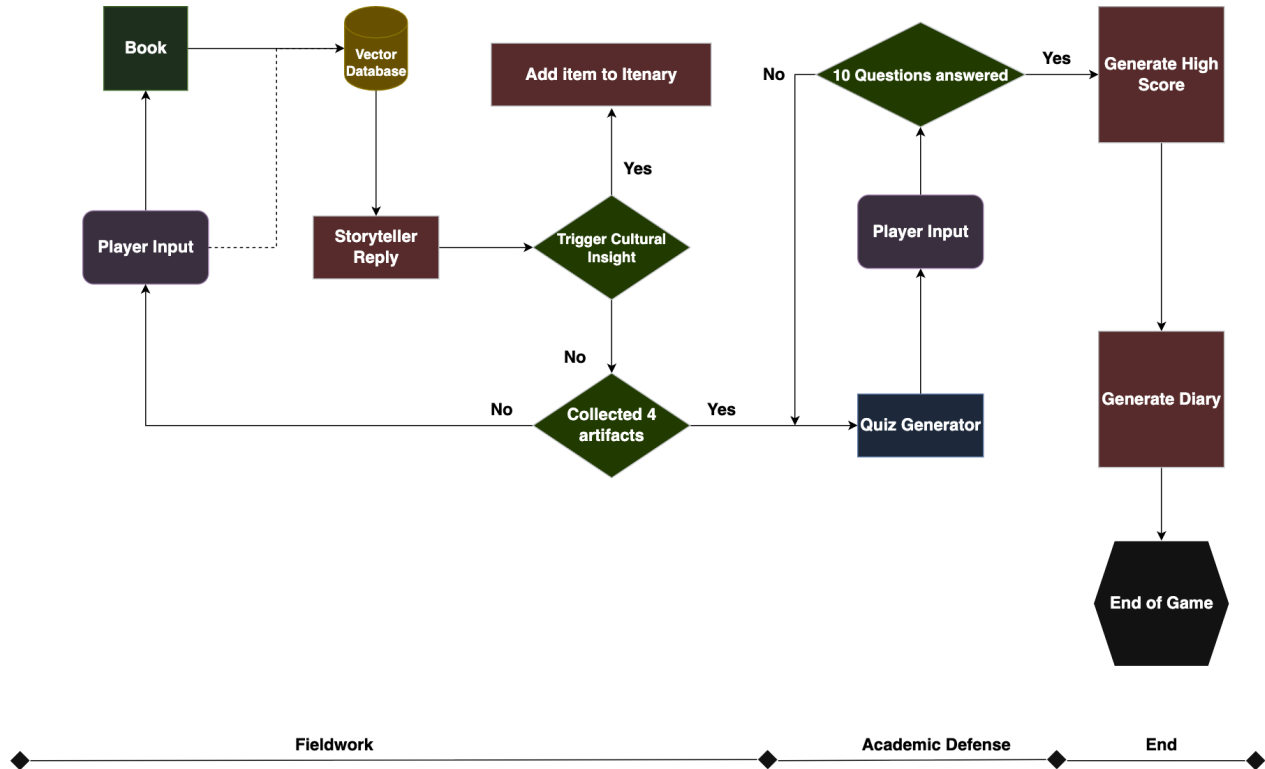


Figure 2: Technical Workflow

The application layer is built on a FastAPI⁶ backend, which manages retrieval orchestration, prompt assembly, and model interaction. The backend exposes endpoints to a React-based single-page frontend built with Vite⁷. The frontend delivers the pixel-based text-adventure interface, handling real-time rendering of images, display of retrieved ethnographic text, and player interaction (including multiple-choice selections and free-text input). Figure 2 illustrates the complete technical workflow, demonstrating the integration between vector-based retrieval, contextual generation, and interactive presentation layers that collectively enable the transformation of classical ethnographic texts into dynamic, educationally-focused gaming experiences⁸.

4 Conclusion and Future Work

Malinowski's Lens demonstrates how a canonical anthropological monograph can be reconfigured as a multimodal, game-based interface through Retrieval-Augmented Generation. Rather than seeking to replace close reading, the system is designed as an interpretive point of entry, sparking curiosity about *Argonauts of the Western Pacific* and motivating further engagement with anthropological texts among audiences with little or no prior exposure to the discipline.

At a technical level, this work illustrates how a RAG-based narrative pipeline can be combined with structured prompting to support

both textual and visual generation while maintaining coherence with a source text. At the level of game design, the collection-based progression foregrounds practices of observation, interpretation, and accumulation, encouraging players to reflect critically on the epistemological and colonial logics embedded in early ethnographic fieldwork. As a Late-Breaking Work, our contribution centers on the design and implementation of the system rather than on formal evaluation.

More broadly, we argue that AI-native, interpretive game interfaces represent a promising design pattern for engaging complex scholarly texts across the humanities and social sciences—particularly in contexts where ambiguity, authorship, and epistemic tension are central to learning rather than obstacles to be resolved.

Future work will include empirical studies with students and community members to examine learning outcomes, interpretive practices, and cultural validity, as well as the development of a no-code platform that enables educators and researchers to transform anthropological texts into interactive, AI-supported learning experiences.

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⁶<https://fastapi.tiangolo.com>

⁷<https://vite.dev>

⁸Although the prototype was inexpensive to develop and test (€33 total), cost-effective manufacturing solutions would be necessary for large-scale classroom implementation

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